

12 (**long**) Consider the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = f(t),$$

with

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \\ 6e^{-t} \cos(t) & \text{for } t \geq 0 \end{cases}$$

and initial conditions $y(0) = y'(0) = 0$.

- (a) Find $\bar{y}(s)$, the Laplace transform of $y(t)$. [10]
- (b) Write $y(t)$ as a convolution between $f(t)$ and another function $g(t)$. State $g(t)$. [10]
- (c) By taking the inverse Laplace transform of $\bar{y}(s)$, find $y(t)$. [10]

END OF PAPER

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Part IA 2019

Paper 4: Mathematical Methods

Sections B and C: Numerical Answers

9. (a) $x(t) = Ae^{-2t} + Be^{-3t}$, for $A, B \in \mathbb{R}$
(b) when $\alpha \neq 2$: $x(t) = Ae^{-2t} + (1 - A)e^{-\alpha t}$, for $A \in \mathbb{R}$
when $\alpha = 2$: $x(t) = (At + 1)e^{-2t}$, for $A \in \mathbb{R}$
(c) $x(t) = Ate^{-2t} + t/4$, for $A \in \mathbb{R}$
10. (a) $A = \begin{bmatrix} \alpha & \frac{1}{2} \\ 1 - \alpha & \frac{1}{2} \end{bmatrix}$
(b) $\mathbf{z}_n \rightarrow \frac{1}{3 - 2\alpha} [1 \ 2(1 - \alpha)]^t$
(c) $\alpha = \frac{1}{2}$
11. (b) $y(t) = \begin{cases} 0 & \text{for } t \leq 0 \\ D(te^{-t} - e^{-t} + e^{-2t}) & \text{for } t > 0 \end{cases}$
(c) $y(t) = \begin{cases} 0 & \text{for } t < T \\ De^{-t}(t - T) - De^{-2t}(e^t - e^T) & \text{for } T \leq t \leq 2T \\ DTe^{-t} - De^{-2t}(e^{2T} - e^T) & \text{for } t > 2T \end{cases}$
12. (a) $\bar{y}(s) = \frac{6(s + 1)}{((s + 1)^2 + 1)^2}$
(b) $g(t) = \begin{cases} 0 & \text{for } t \leq 0 \\ e^{-t} \sin(t) & \text{for } t > 0 \end{cases}$
(c) $y(t) = \begin{cases} 0 & \text{for } t \leq 0 \\ 3te^{-t} \sin(t) & \text{for } t > 0 \end{cases}$